INCH-POUND

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PERFORMANCE SPECIFICATION

NIGHT VISION GOGGLES, AN/PVS-5( )

This specification is approved by the U.S. Army Communications-Electronics Command (CECOM), Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Night Vision Goggles, AN/PVS-5( ) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023, by using the Standardization Document Improvement Proposal (DD Form 1426) or by letter.

AMSC N/A FSC 5855

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SPECIFICATIONS

MILITARY

MIL-PRF-49052 - Image Intensifier Assembly, 18 MM Microchannel Wafer, MX-9916/UV
MIL-PRF-49156 - Lens Assembly, Eyepiece, For Night Vision Goggles, AN/PVS-5( )
MIL-PRF-49157 - Lens Assembly, Objective, For Night Vision Goggles, AN/PVS-5( )

STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics For Equipment, Subsystem And System
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of

2.1.1 Reference documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.b).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-657500 - Night Vision Goggles, AN/PVS-5( ) and all associated drawings
SM-C-657428 - Demisting Shield Assembly
A3144480 - Objective Lens Assembly F/1.05
SM-C-804193 - Eyepiece Assembly
SM-D-657310 - Image Intensifier, Night Vision
SM-D-657337 - Frame Assembly
A3139220 - Mask Assembly
SM-D-657302 - Cushion Assembly, Face
SM-C-657443 - Light Emitting Diode Assembly
SM-D-657430 - Case Assembly, Carrying
SM-D-657440 - Case Assembly, Shipping and Storage
A6169235 - Detector, Highlight Shutdown

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The AN/PVS-5( ) Night Vision Goggles consists of a goggle assembly, a carrying case, a shipping/storage case, and ancillary equipment. The goggle assembly, which is referred to hereinafter as the assembly, is a lightweight, headmounted, self-contained night vision system containing two monocular units comprising a binocular system. Each monocular unit consists of an objective lens assembly, an image intensifier assembly, and an eyepiece assembly. The frame, containing the two monocular units, is mounted to a face mask assembly which is held by head straps to the user's head. The assembly incorporates an infrared (IR) light emitting diode (LED) assembly which provides illumination, when required, to permit close-in-viewing; the assembly also incorporates a highlight shutdown detector which turns off the assembly in a specified time range when exposed to a specified light level range. The assembly permits the user to view at ranges of ten inches to infinity by means of a focus adjustment.

3.2 Construction. The assembly shall be constructed to comply with the performance requirements in the contract, drawing SM-D-657500 as applicable, and as specified herein.

3.3 First article test inspection. When specified in the contract or purchase order, the contractor shall furnish assemblies for first article test (FAT) inspection and approval (see 6.2.e).

3.4 Parts and materials. The contractor shall select parts and materials fully capable of meeting all operational and environmental requirements specified herein. The materials specified on the applicable drawings are recommended, but not mandatory. Verification that the assembly meets the overall performance requirements shall be the governing acceptance standard.

3.5 Major components. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.e).

3.5.1 Demisting shield assembly. The demisting shield assembly shall be in accordance with drawing SM-C-657428 as applicable.

3.5.2 Lens assembly, objective. The objective lens assembly shall be in accordance with MIL-PRF-49157 and drawing A3144480 as applicable.

3.5.3 Lens assembly, eyepiece. The eyepiece lens assembly shall be in accordance with MIL-PRF-49156 and drawing SM-C-804193 as applicable.

3.5.4 Image intensifier assembly. The image intensifier assembly shall be in accordance with MIL-PRF-49052 and drawing SM-D-657310 as applicable.

3.5.5 Frame assembly. The frame assembly shall be in accordance with drawing SM-D-657337 as applicable.

3.5.6 Face Mask assembly. The face mask assembly shall be in accordance with drawing A3139220 as applicable.

3.5.7 Cushion assembly, face. The face cushion assembly shall be in accordance with drawing SM-D-657302 as applicable.

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3.5.8 Light emitting diode assembly. The light emitting diode assembly shall be in accordance with drawing SM-C-657443 as applicable.

3.5.9 Case assembly, carrying. The carrying case assembly shall be in accordance with drawing SM-D-657430 as applicable.

3.5.10 Case assembly, shipping and storage. The shipping and storage case assembly shall be in accordance with drawing SM-D-657440 as applicable.

3.5.11 Detector, highlight shutdown. The highlight shutdown detector shall be in accordance with drawing A3139235 as applicable.

3.6 Performance characteristics.

3.6.1 Controls and adjustments.

3.6.1.1 Rotary switch. The rotary switch shall rotate to 3 distinct positions. In the "OFF" position, neither one of the image intensifier tubes nor the infrared (IR) LED shall light. In the "ON" position, both of the image intensifier tubes shall light and the IR LED shall not light. In the "PULL IR" position, both of the image intensifier tubes shall light and the IR LED shall light.

3.6.1.2 Interpupilary adjustment. With the clamp lever tightened, both monocular assemblies shall be held so as to prevent loss of adjustment. With the clamp lever loosened, monoculars shall be capable of movement throughout their entire interpupilary adjustment range.

3.6.1.3 Relative movement of binocular assembly. The binocular assembly shall be capable of movement relative to the face mask when the two clamp knobs are loosened. The self-locking hex nut shall permit the clamp knob to be loosened while at the same time retaining it on the shouldered screw.

3.6.2 Resolution.

3.6.2.1 Far focus. Each monocular assembly shall have an on-axis resolution at the infinity stop of not less than 20 line pairs per millimeter (lp/mm) or 5 lp/mm less than the maximum tube resolution of the monocular, whichever is greater, and shall have an on-axis resolution at the true infinity setting of not less than 23 lp/mm, or 4 lp/mm less than the maximum tube resolution of each monocular, whichever is greater.

3.6.2.2 Close focus. Each monocular assembly shall have an on-axis resolution at the close focus stop of not less than 23 lp/mm for a target at a distance of not more than 25 centimeters (cm).

3.6.3 Brightness gain. The brightness gain shall not be less than 400 footlamberts per footlambert. There shall be no change in brightness gain greater than 3 percent of the average gain when the assembly is subjected to steady input illumination.

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3.6.4 Collimation. When two, parallel beams of collimated light are projected into the objective lens assemblies, with both objective lens assemblies adjusted to and operating in the infinity focus position, and both eyepiece lens assemblies set at zero diopter, the conjugate beams emerging from the two eyepieces shall be collimated within 1^{o} divergence/convergence in the horizontal plane and within $1/2^{o}$ divergence in the vertical plane.

3.6.5 Focus adjustment. The torque required to adjust the focus range shall be 25 inch-ounces ± 15.

3.6.6 Diopter focus. The minimum diopter focus range shall be from +2 diopters to -6 diopters at a 15 mm eye relief. The torque required to adjust the eyepiece diopter focus shall be 57 inch-ounces ± 47.

3.6.7 Highlight shutdown capability. The highlight shutdown detector shall turn off the assembly within 70 seconds ± 30 when ambient brightness exceeds 4 foot-candles at the photodetector. After the assembly is turned off by the highlight shutdown detector, the rotary switch must be positioned to "OFF" and then "ON" to resume assembly operation.

3.7 Environmental characteristics.

3.7.1 Temperature extremes. The assembly shall not be damaged (see 3.11.1) by storage in any temperature $+23^{o}C$ to $+65^{o}C$ or $+23^{o}C$ to $-51^{o}C$ or by operation at any temperature from $+23^{o}C$ to $+45^{o}C$ or from $+23^{o}C$ to $-51^{o}C$. The assembly shall not be damaged (see 3.11.1) after being subjected to repetitive temperature changes between $+23^{o}C$ to $+65^{o}C$ in a 5 minute period and between $+23^{o}C$ to $-51^{o}C$ in a 5 minute period. Soak time at temperature extremes shall be sufficient for the entire assembly to attain stabilization. Time for temperature stabilization is defined as the amount of time required for the most internal part to achieve the specified temperature. Validation of operation shall be observance of a green glow at the eyepiece at operating temperatures and, at low operating temperature, the ability to operate all controls with arctic gloves or equivalent.

3.7.2 Humidity. The assembly shall not be damaged (see 3.11.1) nor suffer degradation of performance (see 3.11.3) after being subjected to an atmosphere with a relative humidity of 95 percent ± 5 over a temperature range of $20^{o}C$ to $65^{o}C$.

3.7.3 Vibration. The assembly shall operate during, and not be damaged (see 3.11.1) by, vibrations in the frequency range of 10 hertz (Hz) to 55 Hz.

3.7.4 Immersion. The assembly shall not be damaged (see 3.11.1) and shall operate without degradation (see 3.11.3) of performance after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.

3.7.5 Electromagnetic interference. The assembly shall meet the Class A3 requirements of MIL-STD-461 and MIL-STD-462. The required tests are RE02, RS01, RS02, and RS03.

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3.7.6 Interchangeability. Objective lens assemblies and eyepiece lens assemblies shall be physically interchangeable between goggles assemblies without degradation of resolution (see 3.6.2) or diopter focus (see 3.6.6). These interchanges shall be made without modification or adjustment beyond that necessary during replacement of any of the parts or assemblies involved.

3.8 Treatment and painting. Treatment and painting shall be anti-reflective, shall be rust resistant and as specified on the applicable drawing unless otherwise specified in the contract.

3.9 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified In the contract.

3.10 Workmanship. Standards of workmanship shall be such that the assembly will meet all requirements of this specification and any referenced specifications or drawings.

3.11 Technical interpretations. The following technical interpretations are, when referenced in sections 3 or 4, mandatory for this specification.

3.11.1 Damage. Damage is defined as:

a. An electrical failure or malfunction including arcing, corona, intermittent operation, or flickering.

b. Brittleness, cracking, corrosion, failure of finish, dents or breakage of any hardware, or change in binocular alignment.

c. Leakage resulting in condensation, fogging, or moisture on any internal surface.

3.11.2 Ambient (room) temperature. Ambient (room) temperature is defined as $23^{o}C \pm 10\ C^{o}$.

3.11.3 Degradation. Degradation is defined as a significant change in measurable characteristics which results in failure of the assembly to meet specified requirements or indicates that there is an inherent defect in the operating characteristics of the unit.

3.11.4 Environmental temperature. Environmental chamber temperatures shall be controlled within $\pm 5C^{\circ}$ of the specified temperature unless otherwise specified.

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article test inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article test (FAT) inspection. Unless otherwise specified in the contract or purchase order (see 6.2.e), FAT inspection shall be performed by the contractor. Upon completion of FAT inspection, the disposition of the FAT assemblies shall be as specified in the contract or purchase order (see 6.2.f).

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4.2.1 First article testing. Each assembly shall be subjected to the inspections of Table I. Samples selected from the first production lot shall be subjected to the inspections of Table II. Sample size shall be as specified in the contract or purchase order (see 6.2.e). Inspections may be conducted in any order except that the resolution inspection shall be performed last. Failure of any inspection shall be cause for rejection of that assembly and may be cause for failure of the FAT.

Table I. Unit first article test inspection.

Test	Requirement Paragraph
Controls and adjustments	3.6.1
Brightness gain	3.6.3
Collimation	3.6.4
Focus adjustment	3.6.5
Diopter focus	3.6.6
Highlight shutdown capability	3.6.7
Resolution	3.6.2

Table II. Sample first article test inspection.

Inspection	Requirement Paragraph
Temperature extremes	3.7.1
Humidity	3.7.2
Vibration	3.7.3
Immersion	3.7.4
Electro-magnetic interference	3.7.5
Interchangeability	3.7.6
Demisting shield assembly	3.5.1
Resolution	3.6.2

4.2.2 Disposition of FAT inspection samples. After successful completion of FAT inspection, the FAT inspection samples shall be refurbished to the extent necessary and delivered in accordance with the contract or purchase order (see 6.2.f).

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.g). Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.11.2).

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirement shall be as specified in the contract or purchase order (see 6.2.i). When actual packaging of materials is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AN/PVS-5( ) is used for general purpose viewing during periods of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

c. The quantity of demisting shield assemblies, conforming to drawing SM-C-657428, if different than two per AN/PVS-5( ).

d. That the contractor will furnish sufficient BA-3058/U and BA-5567/U batteries for all tests. The batteries used for testing are not to be delivered.

e. Statement identifying that FAT inspection is, or is not required and sampling requirements for FAT.

f. Quantity and schedule for first article testing and disposition of FAT samples.

g. Statement identifying that the contractor must submit his proposal for conformance inspection to include the specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.

h. Necessary actions by the contractor in the event of a lot failure.

i. Levels of preservation and packaging (see 5.0).

6.3 Definitions. See 3.11.

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6.4 Subject/term/keyword listing.

AN/PVS-5( ) night vision goggles
Eyepiece assembly
Objective lens assembly
Image intensifier assembly
Face mask assembly
Carrying case assembly
Shipping and storage case assembly

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodian:
Army - CR
Air Force - 11
Navy - AS

Review Activity
Army - AT

Preparing Activity:
Army - CR

Project 5855-0194

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49065F	**2. DOCUMENT DATE (YYMMDD)** 990304
3. DOCUMENT TITLE Night Vision Goggles, AN/PVS-5()		
4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)		
5. REASON FOR RECOMMENDATION		
6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)
8. PREPARING ACTIVITY		
a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129	
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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